MSML-650 - Cloud Computing - Lab 3 Screenshot

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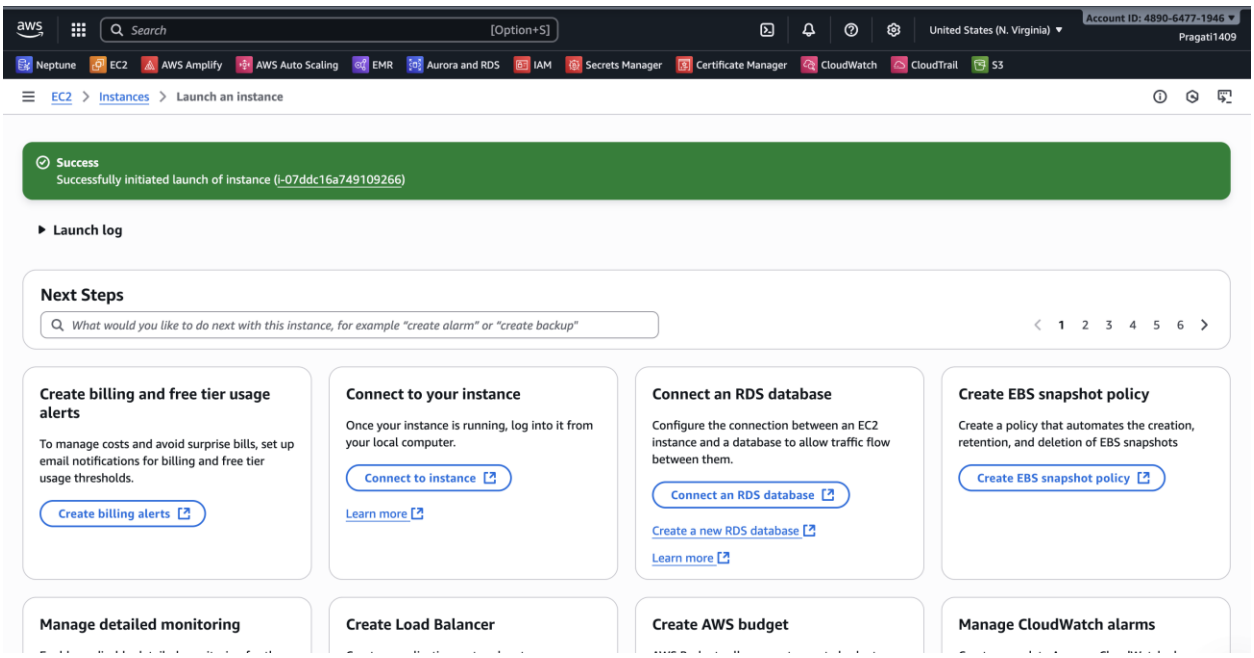
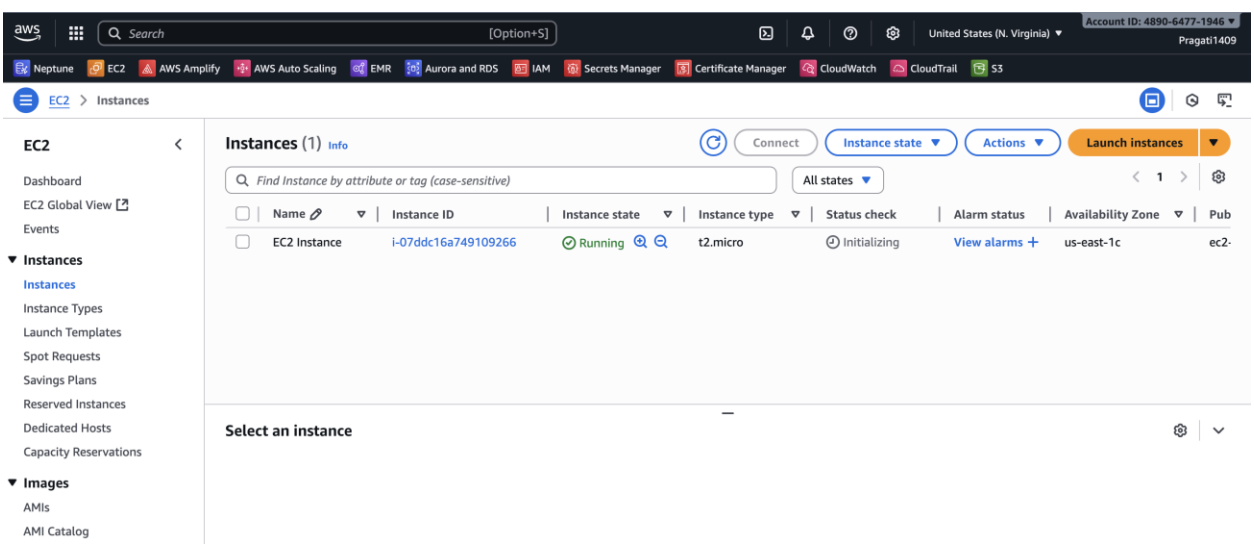
A. Preparation

1. EC2 instance





2. SSH session

```
pragatirao — ec2-user@ip-172-31-31-236:~ — ssh -i ~/.ssh/EC2KeyPair.pem ec2-user...
(base) pragatirao@Pragatis-MacBook-Air-3 ~ % ssh -i ~/.ssh/EC2KeyPair.pem ec2-user@98.81.179.202
The authenticity of host '98.81.179.202 (98.81.179.202)' can't be established.
ED25519 key fingerprint is SHA256:xAy37OU7NaDFqsUOIdiF4LJT7yc+Skftu4h6S6Vyk+k.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '98.81.179.202' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-31-236 ~]$
```

B. Linux commands

3. Linux command ‘df’

```
pragatirao — ec2-user@ip-172-31-31-236:~ — ssh -i ~/.ssh/EC2KeyPair.pem ec2-user...
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system disk space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This manual page documents the GNU version of df.  df displays the amount of disk
       space available on the file system containing each file  name  argument.   If  no
       file  name is given, the space available on all currently mounted file systems is
       shown.  Disk space is shown in 1K blocks by default, unless the environment vari-
       able POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

       If  an  argument  is  the  absolute  file name of a disk device node containing a
       mounted file system, df shows the space available on that file system rather than
       on  the  file  system containing the device node.  This version of df cannot show
       the space available on unmounted file systems, because on most kinds  of  systems
       doing so requires very nonportable intimate knowledge of file system structures.

OPTIONS
       Show  information  about  the file system on which each FILE resides, or all file
 Manual page df(1) line 1 (press h for help or q to quit)
```

4. Install the nano

```
pragatirao — ec2-user@ip-172-31-31-236:~ — ssh -i ~/.ssh/EC2KeyPair.pem ec2-user...
[ec2-user@ip-172-31-31-236 ~]$ man df
[ec2-user@ip-172-31-31-236 ~]$ sudo dnf install -y nano
Amazon Linux 2023 Kernel Livepatch repository              245 kB/s |  26 kB     00:00
Package nano-8.3-1.amzn2023.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-31-236 ~]$
```

5. Create Subdirectory called Course

```
pragatirao — ec2-user@ip-172-31-31-236:~/Course/Backup — ssh -i ~/.ssh/EC2KeyP...
[ec2-user@ip-172-31-31-236 ~]$ mkdir ~/Course
[ec2-user@ip-172-31-31-236 ~]$ mkdir ~/Course/Backup
[ec2-user@ip-172-31-31-236 ~]$ cd ~/Course/Backup
[ec2-user@ip-172-31-31-236 Backup]$ pwd
/home/ec2-user/Course/Backup
[ec2-user@ip-172-31-31-236 Backup]$
```

6. Copy file - /etc/passwd into the Backup directory and rename the file as users.txt

```
pragatirao — ec2-user@ip-172-31-31-236:~/Course/Backup — ssh -i ~/.ssh/EC2KeyP...
[ec2-user@ip-172-31-31-236 Backup]$ cp /etc/passwd .

[ec2-user@ip-172-31-31-236 Backup]$ mv passwd users.txt
[ec2-user@ip-172-31-31-236 Backup]$ ls -l
total 4
-rw-r--r--. 1 ec2-user ec2-user 1519 Oct  4 00:01 users.txt
[ec2-user@ip-172-31-31-236 Backup]$
```

7. Operations on the file users.txt with nano

```
pragatirao — ec2-user@ip-172-31-31-236:~/Course/Backup — ssh -i ~/.ssh/EC2KeyP...
  GNU nano 8.3                          users.txt
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/:/usr/sbin/nologin
                                  [ Read 28 lines ]
^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line
```

```
pragatirao — ec2-user@ip-172-31-31-236:~/Course/Backup — ssh -i ~/.ssh/EC2KeyP...
  GNU nano 8.3                          users.txt                          Modified
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/:/usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/:/usr/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993::/home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin

^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line
```

```
pragatirao — ec2-user@ip-172-31-31-236:~/Course/Backup — ssh -i ~/.ssh/EC2KeyP...
  GNU nano 8.3                          users.txt                          Modified
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/:/usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/:/usr/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993::/home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
ec2-user:x:3456:1000:EC2 Default User:/home/ec2-user:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin

File Name to Write: users.txt
^G Help          M-D DOS Format     M-A Append         M-B Backup File
^C Cancel        M-M Mac Format     M-P Prepend        ^T Browse
```

```
pragatirao — ec2-user@ip-172-31-31-236:~/Course/Backup — ssh -i ~/.ssh/EC2KeyP...
  GNU nano 8.3                          users.txt                          Modified
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/:/usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/:/usr/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993::/home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
ec2-user:x:3456:1000:EC2 Default User:/home/ec2-user:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
There are no more users

^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute     ^C Location
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify     ^/ Go To Line
```

8. Display the content of the file users.txt

```
pragatirao — ec2-user@ip-172-31-31-236:~/Course/Backup — ssh -i ~/.ssh/EC2KeyP...
[ec2-user@ip-172-31-31-236 Backup]$ nano users.txt
[ec2-user@ip-172-31-31-236 Backup]$ nano users.txt
[ec2-user@ip-172-31-31-236 Backup]$ cat users.txt
root:x:0:0:root:/root:/bin/bash
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/:/usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/:/usr/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993::/home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
ec2-user:x:3456:1000:EC2 Default User:/home/ec2-user:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
There are no more users
[ec2-user@ip-172-31-31-236 Backup]$
```

9. Move the file from the Backup subdirectory to your home directory

```
pragatirao — ec2-user@ip-172-31-31-236:~ — ssh -i ~/.ssh/EC2KeyPair.pem ec2-user@98.81.179....
[ec2-user@ip-172-31-31-236 Backup]$ mv users.txt ~/
[ec2-user@ip-172-31-31-236 Backup]$ cd ../..
[ec2-user@ip-172-31-31-236 ~]$ ls -l
total 4
drwxr-xr-x. 3 ec2-user ec2-user   20 Oct  3 23:59 Course
-rw-r--r--. 1 ec2-user ec2-user 1543 Oct  4 00:12 users.txt
[ec2-user@ip-172-31-31-236 ~]$
```

10. List the content of your home directory

Answer:

The **Course directory** has drwxr-xr-x permissions.

- **Owner** can read, write, and enter the directory.
- **Group** and **others** can only read and enter.
  This means only the owner can modify the directory.

The **users.txt file** has -rw-r--r-- permissions.

- **Owner** can read and write.
- **Group** and **others** can only read.
  This means only the owner can edit the file; others can just view it.

11. Remove read and execute privileges

```
pragatirao — ec2-user@ip-172-31-31-236:~ — ssh -i ~/.ssh/EC2KeyPair.pem ec2-user@98.81.179....
[ec2-user@ip-172-31-31-236 ~]$ chmod o-rx ~/Course
[ec2-user@ip-172-31-31-236 ~]$ ls -ld ~/Course
drwxr-x---. 3 ec2-user ec2-user 20 Oct  3 23:59 /home/ec2-user/Course
[ec2-user@ip-172-31-31-236 ~]$
```

12. Add write privileges for users in your primary group

```
pragatirao — ec2-user@ip-172-31-31-236:~ — ssh -i ~/.ssh/EC2KeyPair.pem ec2-user@98.81.179....
[ec2-user@ip-172-31-31-236 ~]$ chmod g+w ~/users.txt
[ec2-user@ip-172-31-31-236 ~]$ ls -l ~/users.txt
-rw-rw-r--. 1 ec2-user ec2-user 1543 Oct  4 00:12 /home/ec2-user/users.txt
[ec2-user@ip-172-31-31-236 ~]$
```

13. prints “Hello World!”

```
pragatirao — ec2-user@ip-172-31-31-236:~/Course — ssh -i ~/.ssh/EC2KeyPair.pem ec2-user@9...
  GNU nano 8.3                                  hello.sh
#! /bin/bash
echo "Hello World!"

                                          [ Read 2 lines ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

14. Save the file as “hello.sh” and exit nano

```
pragatirao — ec2-user@ip-172-31-31-236:~/Course — ssh -i ~/.ssh/EC2KeyPair.pem ec2-user@9...
  GNU nano 8.3                                  hello.sh
#! /bin/bash
echo "Hello World!"

File Name to Write: hello.sh
^G Help              M-D DOS Format       M-A Append           M-B Backup File
^C Cancel            M-M Mac Format       M-P Prepend          ^T Browse
```

15. Try to run the script (type: ./hello.sh). What did you experience? Check the permissions of the file hello.sh. What is the problem?

```
pragatirao — ec2-user@ip-172-31-31-236:~/Course — ssh -i ~/.ssh/EC2KeyPair.pem ec2-user@9...
[ec2-user@ip-172-31-31-236 Course]$ ./hello.sh
-bash: ./hello.sh: Permission denied
[ec2-user@ip-172-31-31-236 Course]$ ls -l hello.sh
-rw-r--r--. 1 ec2-user ec2-user 33 Oct  4 00:26 hello.sh
[ec2-user@ip-172-31-31-236 Course]$
```

Reason: When I tried to run ./hello.sh, the shell returned **"Permission denied."** Checking the file with ls -l hello.sh showed permissions **-rw-r--r--**, which means the file is readable and writable by the owner but **not executable**. Scripts need the execute bit set to run.

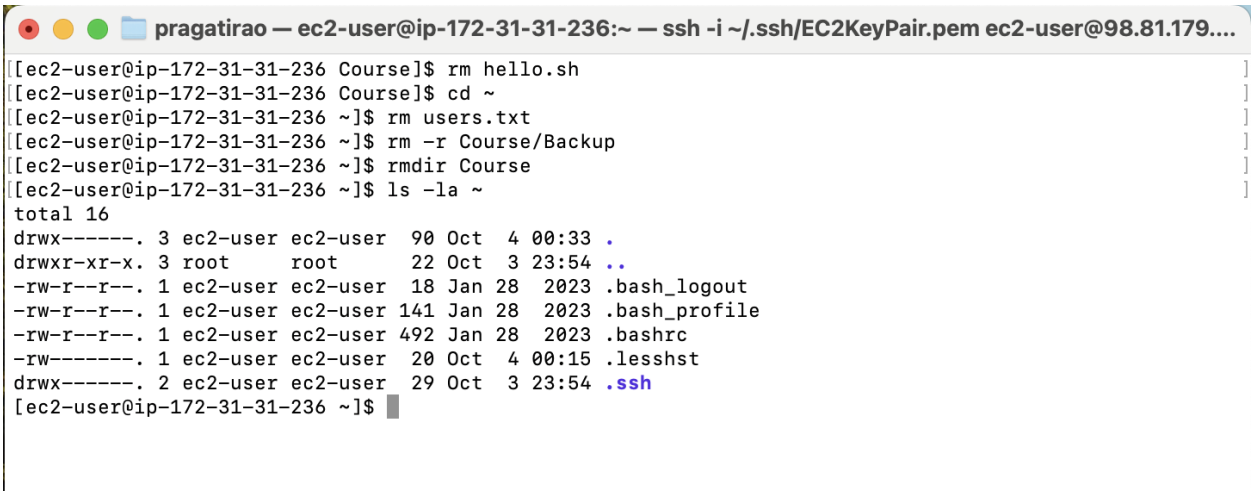
16. Change the file permission

```
pragatirao — ec2-user@ip-172-31-31-236:~/Course — ssh -i ~/.ssh/EC2KeyPair.pem ec2-user@9...
[ec2-user@ip-172-31-31-236 Course]$ chmod u+x hello.sh
[ec2-user@ip-172-31-31-236 Course]$ ls -l hello.sh
-rwxr--r--. 1 ec2-user ec2-user 33 Oct  4 00:26 hello.sh
[ec2-user@ip-172-31-31-236 Course]$
```

17. Try to run the script again. Now you should see the printout on the screen

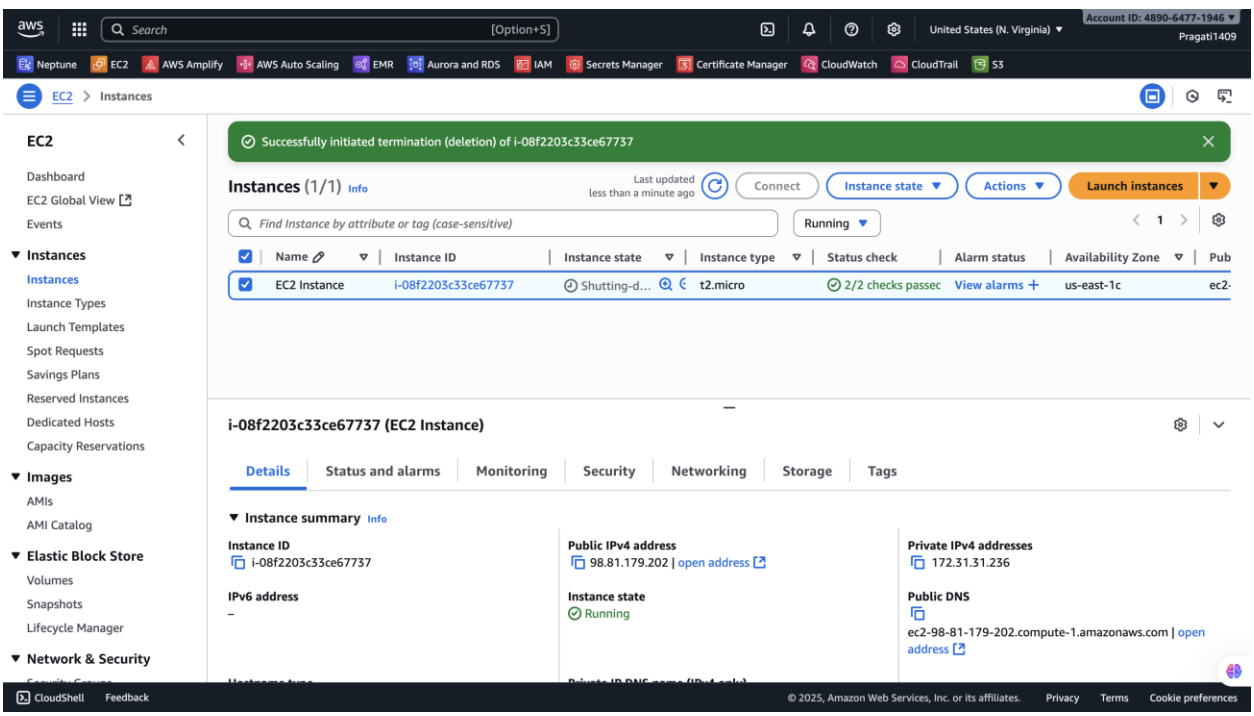
```
pragatirao — ec2-user@ip-172-31-31-236:~/Course — ssh -i ~/.ssh/EC2KeyPair.pem ec2-user@9...
[ec2-user@ip-172-31-31-236 Course]$ ./hello.sh
Hello World!
[ec2-user@ip-172-31-31-236 Course]$
```

18. Delete the file hello.sh.



C. Clean up after yourself

19. Terminate the EC2 instance



20. Log out -> Done